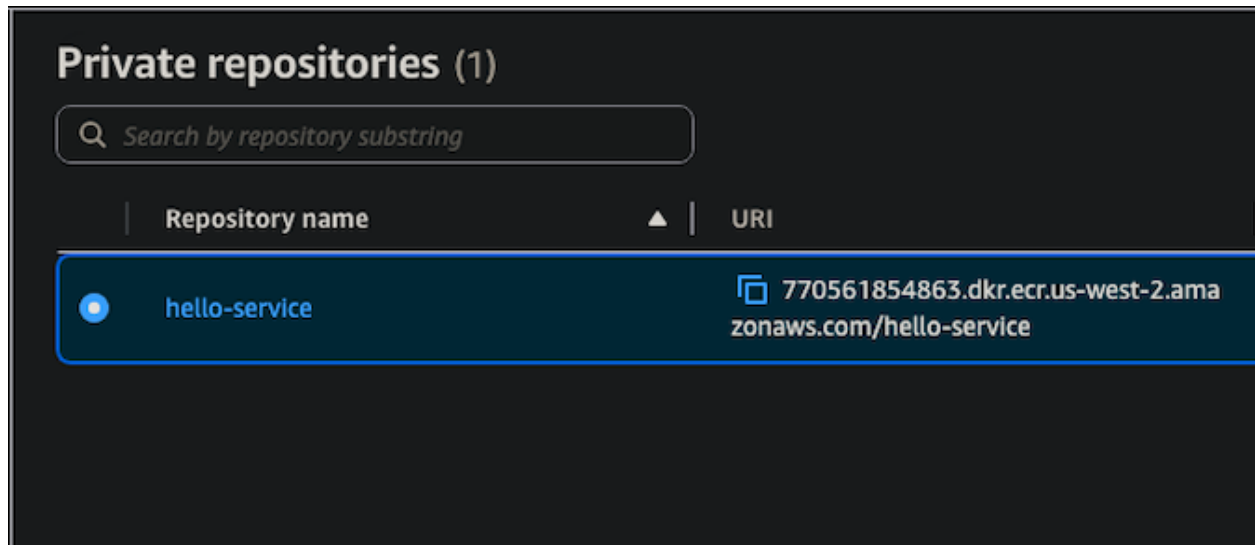


Step 2 - ECR:



Step 3 - Verify ECR Images

```
(base) pranaybeela@Pranays-MacBook-Pro hw4 % aws ecr list-images \
  --repository-name hello-service \
  --region us-west-2 \
  --query 'imageIds[*].imageTag' \
  --output table
-----
|ListImages|
+-----+
|  latest  |
+-----+
(base) pranaybeela@Pranays-MacBook-Pro hw4 %
```

Step 4 - ECS Cluster

✓ Tasks launched

- arn:aws:ecs:us-west-2:770561854863:task/hello-cluster-1/3c9e1924ef1e4f58b0b61d7e96efd014

Notifications 0

hello-cluster-1

Cluster overview

ARN	arn:aws:ecs:us-west-2:770561854863:cluster/hello-cluster-1	Status	✓ Active
Services			
Draining	-	Active	-

Services

Tasks

Infrastructure

Metrics

Scheduled tasks

Configuration

Step 5 - Task Definition

✓ Task definition successfully created
hello-task:2 has been successfully created. You can use this task definition to deploy a service or run a task.

Task definitions (4) [Info](#)

Last updated 8 February 2026, 19:42 (UTC-5:00)

Deploy ▼

Filter task definitions

Filter status Active

Task definition	Status of last rev
☐ hello-task	✓ Active

Step 6 - Running a Task

The screenshot shows the Amazon Elastic Container Service console. The breadcrumb navigation at the top reads: Amazon Elastic Container Service > Clusters > hello-cluster-1 > Tasks > 3c9e1924ef1e4f58b0b61d7e96efd014 > Configuration. On the left sidebar, the 'Clusters' section is expanded, showing options like Express Mode, Clusters, Namespaces, Task definitions, Account settings, Amazon ECR, Repositories, and AWS Batch. The main content area has a green header bar with a checkmark icon and the text 'Tasks launched', followed by the task ARN: arn:aws:ecs:us-west-2:770561854863:task/hello-cluster-1/3c9e1924ef1e4f58b0b61d7e96efd014. Below this is a 'Notifications' button. The task ID '3c9e1924ef1e4f58b0b61d7e96efd014' is displayed in large text. A tab bar below the ID includes 'Configuration' (selected), 'Metrics', 'Logs', 'Networking', 'Volumes (0)', and 'Tags'. The 'Task overview' section shows the ARN and a 'Last status' of 'Running' with a green checkmark icon.

Amazon Elastic Container Service

Express Mode

Clusters

Namespaces

Task definitions

Account settings

Amazon ECR

Repositories

AWS Batch

Tasks launched

arn:aws:ecs:us-west-2:770561854863:task/hello-cluster-1/3c9e1924ef1e4f58b0b61d7e96efd014

Notifications

3c9e1924ef1e4f58b0b61d7e96efd014

Configuration | Metrics | Logs | Networking | Volumes (0) | Tags

Task overview

ARN

arn:aws:ecs:us-west-2:770561854863:task/hello-cluster-1/3c9e1924ef1e4f58b0b61d7e96efd014

Last status

Running

Step 7 - Public IP and Testing

This screenshot shows the 'Networking' tab for the same task. The breadcrumb navigation is identical to the previous screenshot. The main content area shows the task ID '3c9e1924ef1e4f58b0b61d7e96efd014' and a date '5 February'. The 'Networking' tab is selected in the tab bar. The 'Network' section displays several details: ENI ID (eni-02a2ce7130d2891d1), Subnet (subnet-0f159122bd5751a2e), Security groups (sg-005e3da7deade5a30), Task role (LabRole), Task execution role (LabRole), Public IP (54.202.224.38), Private IP (172.31.7.137), IPv6 address (-), and MAC address (0a:25:25:73:d3:e7). Each detail is accompanied by a small icon and a link to view more information.

Tasks launched

arn:aws:ecs:us-west-2:770561854863:task/hello-cluster-1/3c9e1924ef1e4f58b0b61d7e96efd014

Notifications

3c9e1924ef1e4f58b0b61d7e96efd014

5 February

Configuration | Metrics | Logs | Networking | Volumes (0) | Tags

Network

ENI ID

eni-02a2ce7130d2891d1

Subnet

subnet-0f159122bd5751a2e

Security groups

sg-005e3da7deade5a30 (hello-service-sg)

Task role

LabRole

Task execution role

LabRole

Public IP

54.202.224.38 | open address

Private IP

172.31.7.137

IPv6 address

-

MAC address

0a:25:25:73:d3:e7

```

[(base) pranaybeela@Pranays-MacBook-Pro hw2 % curl http://54.202.224.38:8080/albums
[
  {
    "id": "1",
    "title": "Blue Train",
    "artist": "John Coltrane",
    "price": 56.99
  },
  {
    "id": "2",
    "title": "Jeru",
    "artist": "Gerry Mulligan",
    "price": 17.99
  },
  {
    "id": "3",
    "title": "Sarah Vaughan and Clifford Brown",
    "artist": "Sarah Vaughan",
    "price": 39.99
  }
]
(base) pranaybeela@Pranays-MacBook-Pro hw2 %

```

```

[(base) pranaybeela@Pranays-MacBook-Pro hw2 % curl http://54.202.224.38:8080/albums/1
{
  "id": "1",
  "title": "Blue Train",
  "artist": "John Coltrane",
  "price": 56.99
}
(base) pranaybeela@Pranays-MacBook-Pro hw2 %

```

Difference in choosing EC2 vs ECS

- EC2 is a virtual machine. We manage the server, OS, updates, scaling, and deployments.
- ECS runs containers for us. We manage task definitions and scaling rules, and AWS handles scheduling and restarts. With Fargate, we do not manage servers at all.

What is a VPC and subnet? How did you get access to the default VPC?

- A VPC is our private network space inside AWS.
- A subnet is a smaller network segment inside a VPC, usually tied to one availability zone. Public subnets route to the internet, private subnets do not.
- We got access by selecting the default VPC that AWS creates for the account and using its default subnets and route tables. Our task got a public IP and used the VPC internet gateway.

What is TCP? How is it different from UDP?

- TCP is connection-based and reliable. It guarantees ordered delivery and retransmits lost packets.
- UDP is connectionless and faster but not reliable. It does not guarantee delivery or ordering.

How do you control resources allocated to a task?

- In the ECS task definition, set CPU and memory for the task and optionally for each container.
- If we use Fargate, we pick a valid CPU and memory combination and AWS enforces it.

Part 3 - Results & Screenshots

Build Container Images for Splitter, Mapper, and Reducer

```
((base) pranaybeela@Pranays-MacBook-Pro hw4 % docker tag splitter:latest 770561854863.dkr.ecr.us-west-2.amazonaws.com/splitter:latest
(base) pranaybeela@Pranays-MacBook-Pro hw4 % docker push 770561854863.dkr.ecr.us-west-2.amazonaws.com/splitter:latest
The push refers to repository [770561854863.dkr.ecr.us-west-2.amazonaws.com/splitter]
c37d8adaec42: Pushed
7cc028d86d16: Pushed
017886f7e176: Pushed
2780920e5dbf: Pushed
52630fc75a18: Pushed
253e490c8ad8: Pushed
62de241dac5f: Pushed
069d1e267530: Pushed
7c12895b777b: Pushed
3214acf345c0: Pushed
bfb59b82a9b6: Pushed
d1c559a043f5: Pushed
cf7c1f75ddb0: Pushed
dcaa5a89b0cc: Pushed
dd64bf2dd177: Pushed
4aa0ea1413d3: Pushed
latest: digest: sha256:c0fd9d6272f6e6bacb5dfa63170e4942282e7d23e9e61f4a0a0e18506903ae size: 856
(base) pranaybeela@Pranays-MacBook-Pro hw4 % █
```

```
((base) pranaybeela@Pranays-MacBook-Pro hw4 % docker tag mapper:latest 770561854863.dkr.ecr.us-west-2.amazonaws.com/mapper:latest
(base) pranaybeela@Pranays-MacBook-Pro hw4 % docker push 770561854863.dkr.ecr.us-west-2.amazonaws.com/mapper:latest
The push refers to repository [770561854863.dkr.ecr.us-west-2.amazonaws.com/mapper]
7cc028d86d16: Pushed
bfb59b82a9b6: Pushed
d1c559a043f5: Pushed
2780920e5dbf: Pushed
3214acf345c0: Pushed
7c12895b777b: Pushed
dcaa5a89b0cc: Pushed
77b371c9251f: Pushed
dd64bf2dd177: Pushed
4aa0ea1413d3: Pushed
253e490c8ad8: Pushed
017886f7e176: Pushed
62de241dac5f: Pushed
52630fc75a18: Pushed
069d1e267530: Pushed
9fec243a165b: Pushed
latest: digest: sha256:b296af888d5d6f659afbe2d6d88d2e225d33862d806265617994e5bed9fc9c62 size: 856
(base) pranaybeela@Pranays-MacBook-Pro hw4 % █
```

```
((base) pranaybeela@Pranays-MacBook-Pro hw4 % docker push 770561854863.dkr.ecr.us-west-2.amazonaws.com/reducer:latest
The push refers to repository [770561854863.dkr.ecr.us-west-2.amazonaws.com/reducer]
4aa0ea1413d3: Pushed
7c12895b777b: Pushed
52630fc75a18: Pushed
dd64bf2dd177: Pushed
253e490c8ad8: Pushed
7cc028d86d16: Pushed
017886f7e176: Pushed
069d1e267530: Pushed
aab982f80308: Pushed
dcaa5a89b0cc: Pushed
64832648c98c: Pushed
2780920e5dbf: Pushed
62de241dac5f: Pushed
3214acf345c0: Pushed
d1c559a043f5: Pushed
bfb59b82a9b6: Pushed
latest: digest: sha256:cd5c01499e49809f91c06529dca973a4ea7cb4d0fca31f62a66ac5e5ef9de429 size: 856
(base) pranaybeela@Pranays-MacBook-Pro hw4 % █
```

Create ECR Repositories for Splitter, Mapper, and Reducer

```
[(base) pranaybeela@Pranays-MacBook-Pro hw4 % aws ecr create-repository --repository-name splitter]
--region us-west-2
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-west-2:770561854863:repository/splitter",
    "registryId": "770561854863",
    "repositoryName": "splitter",
    "repositoryUri": "770561854863.dkr.ecr.us-west-2.amazonaws.com/splitter",
    "createdAt": 1770347954.592,
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}
```

```
[(base) pranaybeela@Pranays-MacBook-Pro hw4 % aws ecr create-repository --repository-name mapper ]
--region us-west-2
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-west-2:770561854863:repository/mapper",
    "registryId": "770561854863",
    "repositoryName": "mapper",
    "repositoryUri": "770561854863.dkr.ecr.us-west-2.amazonaws.com/mapper",
    "createdAt": 1770347964.898,
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}
```

```
[(base) pranaybeela@Pranays-MacBook-Pro hw4 % aws ecr create-repository --repository-name reducer]
--region us-west-2
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-west-2:770561854863:repository/reducer",
    "registryId": "770561854863",
    "repositoryName": "reducer",
    "repositoryUri": "770561854863.dkr.ecr.us-west-2.amazonaws.com/reducer",
    "createdAt": 1770347972.938,
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}
```

Private repositories (4)

View push commands

Delete

Actions

Create repository

Search by repository substring

	Repository name	URI	Created at	Tag immutability	Encryption type
	hello-service	770561854863.dkr.ecr.us-west-2.amazonaws.com/hello-service	05 February 2026, 20:03:43 (UTC-05)	Immutable	AES-256
	mapper	770561854863.dkr.ecr.us-west-2.amazonaws.com/mapper	05 February 2026, 22:19:24 (UTC-05)	Immutable	AES-256
	reducer	770561854863.dkr.ecr.us-west-2.amazonaws.com/reducer	05 February 2026, 22:19:32 (UTC-05)	Immutable	AES-256
	splitter	770561854863.dkr.ecr.us-west-2.amazonaws.com/splitter	05 February 2026, 22:19:14 (UTC-05)	Immutable	AES-256

Create ECS Task Definitions and Tasks for Splitter, Mapper, and Reducer

Task definitions (4)

Info

Last updated
8 February 2026, 20:09 (UTC-5:00)

Deploy

Create new revision

Create new task definition

Filter task definitions

Filter status
Active

< 1 >

⚙

Task definition	Status of last revision
☐ hello-task	✔ Active
☐ mapper-task	✔ Active
☐ reducer-task	✔ Active
☐ splitter-task	✔ Active

ServicesTasksInfrastructureMetricsScheduled tasksConfigurationEvent historyTags

Tasks (7)

Last updated
8 February 2026, 18:58 (UTC-5:00)

Manage tags

Stop

Run new task

Filter tasks by property or value

Filter desired status
Any desired status

Filter launch type
Any launch type

< 1 >

⚙

☐	Task	Last status	Desired status	Task definition	Health status	Created at
☐	 13fb659faa5d4ff4a92090...	✔ Running	✔ Running	mapper-task:1	ⓘ Unknown	6 minutes ago
☐	 ac8f6eb02cb344789955d...	✔ Running	✔ Running	splitter-task:1	ⓘ Unknown	8 minutes ago
☐	 c8e4bc3241ec4cd2a80cf...	✔ Running	✔ Running	mapper-task:1	ⓘ Unknown	7 minutes ago
☐	 f56bdf2c6c1e49bd82e8a...	✔ Running	✔ Running	mapper-task:1	ⓘ Unknown	6 minutes ago
☐	 fe541d99ce0f40588d52d...	✔ Running	✔ Running	reducer-task:1	ⓘ Unknown	5 minutes ago
☐	 6a29fe74ff114cbc9bed30...	⛔ Stopped ...	⊖ Stopped	splitter-task:1	ⓘ Unknown	23 minutes ago
☐	 bf2c7f3e936e4167918be...	⛔ Stopped ...	⊖ Stopped	mapper-task:1	ⓘ Unknown	20 minutes ago

Run Splitter Task and Generate S3 Chunk URLs

```
((base) pranaybeela@Pranays-MacBook-Pro hw4 % curl "http://54.202.48.113:8080/split?url=https://raw.githubusercontent.com/teropa/nlp/master/resources/corpora/gutenberg/shakespeare-hamlet.txt"
{"chunks":["s3://amzn-bucket-text-file-12345678/chunks/1770595485/chunk1.txt","s3://amzn-bucket-text-file-12345678/chunks/1770595485/chunk2.txt","s3://amzn-bucket-text-file-12345678/chunks/1770595485/chunk3.txt"]}
(base) pranaybeela@Pranays-MacBook-Pro hw4 %
```

Run Mapper Tasks and Generate Intermediate JSON Outputs

```
((base) pranaybeela@Pranays-MacBook-Pro hw4 % curl "http://44.250.172.177:8080/map?chunk=s3://amzn-bucket-text-file-12345678/chunks/1770595485/chunk1.txt"
{"out":"s3://amzn-bucket-text-file-12345678/maps/1770595573/chunk1.json"}
(base) pranaybeela@Pranays-MacBook-Pro hw4 % curl "http://54.149.23.179:8080/map?chunk=s3://amzn-bucket-text-file-12345678/chunks/1770595485/chunk2.txt"
{"out":"s3://amzn-bucket-text-file-12345678/maps/1770595613/chunk2.json"}
(base) pranaybeela@Pranays-MacBook-Pro hw4 % curl "http://44.244.152.60:8080/map?chunk=s3://amzn-bucket-text-file-12345678/chunks/1770595485/chunk3.txt"
{"out":"s3://amzn-bucket-text-file-12345678/maps/1770595641/chunk3.json"}
(base) pranaybeela@Pranays-MacBook-Pro hw4 %
```

Run Reducer Task and Generate Final Output JSON

```
((base) pranaybeela@Pranays-MacBook-Pro hw4 % curl "http://44.249.190.218:8082/reduce?m1=s3://amzn-bucket-text-file-12345678/maps/1770595573/chunk1.json&m2=s3://amzn-bucket-text-file-12345678/maps/1770595613/chunk2.json&m3=s3://amzn-bucket-text-file-12345678/maps/1770595641/chunk3.json"
{"out":"s3://amzn-bucket-text-file-12345678/reduce/1770595726/final.json"}
(base) pranaybeela@Pranays-MacBook-Pro hw4 %
```

Verify S3 Outputs in the Console

1770344441/

Objects | Properties

Objects (3)

Copy S3 URI

Copy URL

Download

Open ↗

Delete

Actions ▼

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	chunk1.txt	txt	February 5, 2026, 21:20:42 (UTC-05:00)	
☐	chunk2.txt	txt	February 5, 2026, 21:20:42 (UTC-05:00)	
☐	chunk3.txt	txt	February 5, 2026, 21:20:43 (UTC-05:00)	

1770595485/

Objects | Properties

Objects (3)

Copy S3 URI

Copy URL

Download

Open ↗

Delete

Actions ▼

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	chunk1.txt	txt	February 8, 2026, 19:04:46 (UTC-05:00)	
☐	chunk2.txt	txt	February 8, 2026, 19:04:46 (UTC-05:00)	
☐	chunk3.txt	txt	February 8, 2026, 19:04:46 (UTC-05:00)	

Objects (6)



Copy S3 URI

Copy URL

Download

Open ↗

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) need to explicitly grant them permissions. [Learn more ↗](#)

 Find objects by prefix

☐	Name	▲	Type	▼	Last modified
☐	 1770345257/		Folder		-
☐	 1770345318/		Folder		-
☐	 1770345328/		Folder		-
☐	 1770595573/		Folder		-
☐	 1770595613/		Folder		-
☐	 1770595641/		Folder		-

1770345513/

Objects

Properties

Objects (1)



Copy S3 URI



Copy URL



Download

Objects are the fundamental entities stored in Amazon S3. You can use [Ar](#) need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix



Name



Type



[final.json](#)

json

Verifying the Counts

```
(base) pranaybeela@Pranays-MacBook-Pro hw4 % aws s3 cp s3://amzn-bucket-text-file-12345678/reduce/1770595726/final.json - \
| python3 -c 'import sys,json; d=json.load(sys.stdin); print("hamlet:", d.get("hamlet"), "the:", d.get("the"))'

hamlet: 100 the: 993
(base) pranaybeela@Pranays-MacBook-Pro hw4 % curl -s "https://raw.githubusercontent.com/teropa/nlp/master/resources/corpora/gutenberg/s
hakespeare-hamlet.txt" \
| tr -cs "A-Za-z" "\n" | tr 'A-Z' 'a-z' \
| awk 'BEGIN{h=0;t=0} $0=="hamlet"{h++} $0=="the"{t++} END{print "hamlet:",h,"the:",t}'

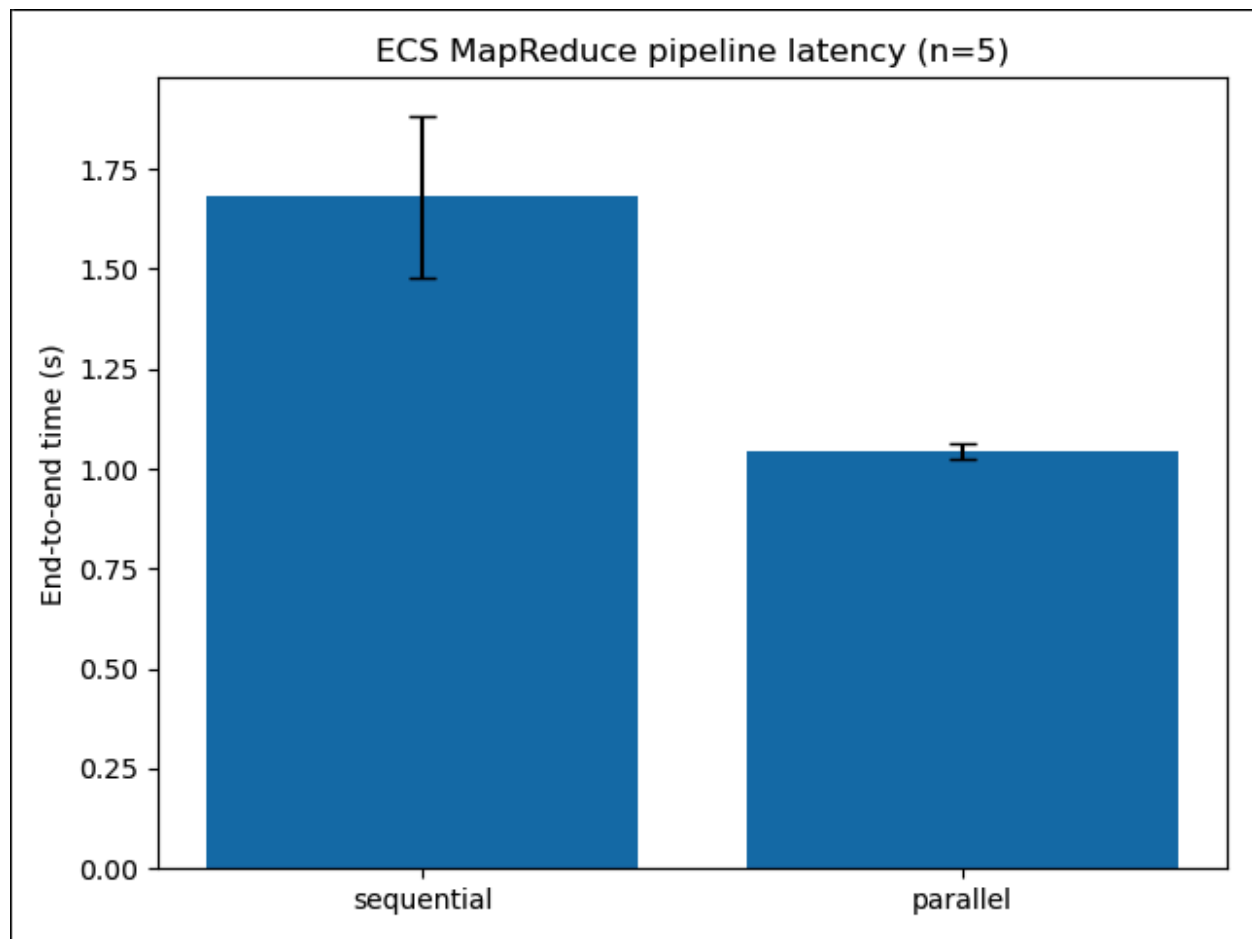
hamlet: 100 the: 993
(base) pranaybeela@Pranays-MacBook-Pro hw4 %
```

Comparing the Performance

```
(base) pranaybeela@Pranays-MacBook-Pro hw4 % export INPUT_URL="https://raw.githubusercontent.com/teropa/nlp/master/resources/corpora/gutenberg/shakespeare-hamlet.txt"

(base) pranaybeela@Pranays-MacBook-Pro hw4 % python3 plot.py
sequential-mappers run 1/5: 2.046s
sequential-mappers run 2/5: 1.641s
sequential-mappers run 3/5: 1.707s
sequential-mappers run 4/5: 1.572s
sequential-mappers run 5/5: 1.440s
parallel-mappers run 1/5: 1.058s
parallel-mappers run 2/5: 1.008s
parallel-mappers run 3/5: 1.063s
parallel-mappers run 4/5: 1.033s
parallel-mappers run 5/5: 1.059s

Summary (seconds):
sequential: mean=1.681 std=0.203
parallel: mean=1.044 std=0.021
```



What do the times and the graph say?

Parallel is faster.

- Sequential pipeline averaged 1.681 s with higher variability (std 0.203 s).
- Parallel pipeline averaged 1.044 s and was very consistent (std 0.021 s).
- Speedup is about 1.6x because the three mappers run at the same time, so total time is closer to the slowest mapper instead of the sum of all three.